

Homework2

February 4, 2019

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In [1]: import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
from scipy.stats import norm
import sklearn.linear_model
from sklearn.linear_model import Lasso

import pandas as pd
import sklearn

from sklearn.linear_model import lasso_path, enet_path
from itertools import cycle

import matplotlib.pyplot as plt

In [2]: import warnings
warnings.filterwarnings('ignore')
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1 Problem 1: Prediction error and dimension.

Use regular linear regression, average squared prediction errors increase with p , while with Lasso regression, average squared prediction errors remain almost the same, which is approximately one.

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In [5]: def RV_generator(p):
    n = 100
    X = np.random.randn(n, p)
    Y = X[:, 0] * 1/np.sqrt(2) + np.random.randn(n)/np.sqrt(2)
    return X, Y

avg_mse = []
avg_mse_lasso = []
for p in range(2,81):
    mse = 0
    mse_lasso = 0
    for i in range(100):
        X,Y = RV_generator(p)
```

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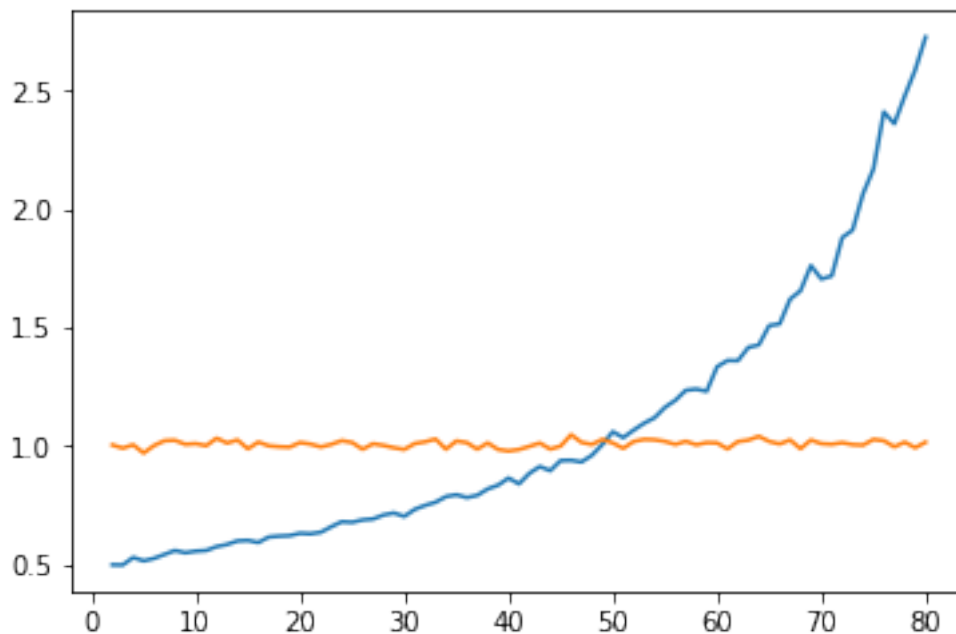
X_test,Y_test = RV_generator(p)
model = sklearn.linear_model.LinearRegression()
model.fit(X,Y)
Y_pred = model.predict(X_test)
mse += sklearn.metrics.mean_squared_error(Y_test, Y_pred) / 100

model_lasso = Lasso(alpha = np.sqrt(2*np.log(p)/100), normalize=True, fit_intercept=True)
model_lasso.fit(X, Y)
# Then call model.predict, etc.
Y_pred_lasso = model_lasso.predict(X_test)
mse_lasso += sklearn.metrics.mean_squared_error(Y_test, Y_pred_lasso) / 100

avg_mse.append(mse)
avg_mse_lasso.append(mse_lasso)

plt.plot(range(2,81),avg_mse)
plt.plot(range(2,81),avg_mse_lasso)
plt.show()

```



In []: